

Discrete optimization

Modeling

Michel Bierlaire

Optimization: principles and algorithms



ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

Modeling

Binary variables are convenient to model many situations.

Action to be taken or not.

A switch to set to “on”

In this video, we investigate some techniques to translate logical rules into a mathematical formulation involving binary variables.

Logical identity

$$x = \begin{cases} 1 & \text{if } P \text{ is true,} \\ 0 & \text{if } P \text{ is false.} \end{cases}$$

P	x
True	1
False	0

Logical negation

$$P : x$$
$$\neg P : 1 - x$$

P	$\neg P$	x	$1 - x$
True	False	1	0
False	True	0	1

Logical conjunction

$$P : x$$

$$Q : y$$

$$P \wedge Q : xy$$

P	Q	$P \wedge Q$	x	y	xy
True	True	True	1	1	1
True	False	False	1	0	0
False	True	False	0	1	0
False	False	False	0	0	0

*If both variables, non linear formulation. Combination of two constraints instead.
Objective here: be comprehensive.*

Logical disjunction

$P : x$

$Q : y$

$P \vee Q : x + y \geq 1$

P	Q	$P \vee Q$	x	y	$x + y \geq 1$
True	True	True	1	1	Yes
True	False	True	1	0	Yes
False	True	True	0	1	Yes
False	False	False	0	0	No

Generalization: $P_1 \vee \dots \vee P_r. \sum_{i=1}^r x_i \geq 1.$

Logical exclusive disjunction

$P : x$

$Q : y$

$P \oplus Q : x + y = 1$

P	Q	$P \oplus Q$	x	y	$x + y = 1$
True	True	False	1	1	No
True	False	True	1	0	Yes
False	True	True	0	1	Yes
False	False	False	0	0	No

Logical implication

$P : x$

$Q : y$

$P \Rightarrow Q : x \leq y$

P	Q	$P \Rightarrow Q$	x	y	$x \leq y$
True	True	True	1	1	Yes
True	False	False	1	0	No
False	True	True	0	1	Yes
False	False	True	0	0	Yes

Logical equivalence

$P : x$

$Q : y$

$P \Leftrightarrow Q : x = y$

P	Q	$P \Leftrightarrow Q$	x	y	$x = y$
True	True	True	1	1	Yes
True	False	False	1	0	No
False	True	False	0	1	No
False	False	True	0	0	Yes

Optional constraint $\geq$

- ▶ z is a binary variable.
- ▶ If $z = 1$, the constraint $f(x) \geq a$ must be verified.
- ▶ If $z = 0$, the constraint $f(x) \geq a$ must not be verified.

Assumption: f is bounded from below by L .

$f(x) - L \geq 0$ is always true.

$$f(x) \geq L + (a - L)z$$

Optional constraint $\leq$

- ▶ z is a binary variable.
- ▶ If $z = 1$, the constraint $f(x) \leq a$ must be verified.
- ▶ If $z = 0$, the constraint $f(x) \leq a$ must not be verified.

Assumption: f is bounded from above by M .

$f(x) \leq M$ is always true.

$$f(x) \leq az + (1 - z)M$$

Disjunctive constraints

- ▶ Constraint 1: $f(x) \geq a$.
- ▶ Constraint 2: $g(x) \geq b$.
- ▶ One of them must be verified, but not necessarily both.

Assumption: f and g are bounded from below.

$f(x) \geq L_f$ and $g(x) \geq L_g$ are always true.

Introduce a binary variable z

$$f(x) \geq L_f + (a - L_f)z$$

$$g(x) \geq L_g + (b - L_g)(1 - z)$$

Linearization

Non linear specification

$$xy = z, x, y, z \in \{0, 1\}.$$

$$x + y \leq 1 + z$$

$$z \leq x$$

$$z \leq y.$$

x	y	z	$x + y \leq 1 + z$	$z \leq x$	$z \leq y$	$xy = z$
1	1	1	Yes	Yes	Yes	Yes
1	1	0	No	Yes	Yes	No
1	0	1	Yes	Yes	No	No
1	0	0	Yes	Yes	Yes	Yes
0	1	1	Yes	No	Yes	No
0	1	0	Yes	Yes	Yes	Yes
0	0	1	Yes	No	No	No
0	0	0	Yes	Yes	Yes	Yes

Summary

*A binary variable can be used to represent a logical statement
In this video, we have reviewed how to translate various logical rules into
mathematical formulations that can be included in an optimization problem.*